

Andrew Barchuk

SOFTWARE ENGINEER

Berlin, Germany

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Summary

A software engineer with more than 13 years of professional experience focusing on building platforms. The areas of focus include *distributed systems, concurrency, scalability and performance, monitoring and observability*. Passionate about building better systems, deepening our understanding of technology, optimizing software delivery and developer productivity.

Skills and technologies

programming	Java, Rust, Scala, Go, C++, Python, JavaScript, Bash
data storage	Kafka, ZooKeeper, Cassandra, Spanner, PostgreSQL, MongoDB
software design	software architecture and design patterns, microservices, CQRS · algorithm analysis and implementation, concurrency
web	HTTP and design of RESTful APIs, WebSockets, understanding of underlying networking stack (TCP, UDP, DNS)
performance	profiling and optimization: heap analysis, CPU sampling, JVM GC, benchmarking, load testing
testing	TDD practitioner, JUnit, Mockito, property-based testing
DevOps	Kubernetes, Docker · AWS, GCP · Prometheus, Grafana, distributed tracing · Linux/Unix administration, ZFS, RAID
security	securing APIs, OAuth, applied cryptography
languages	English, German (B2), French (A1), Ukrainian, Russian

Experience

Zalando

Berlin, Germany

PRINCIPAL SOFTWARE ENGINEER, RISK AND ABUSE MANAGEMENT

Oct. 2022 - Present

- Lead a multi-year effort to build a near-real-time fraud detection platform. Define a foundation for MLOps risk infrastructure.
- Design the Zalando invitation-only purchase model and lead the data migration to an event-sourced order storage. Contribute to the design of the central ML & AI platform. Facilitate adoption of Apache Flink as the company-standard stream processing framework.
- Drive technology vision, target architecture, and technology stack of a three-team department.

LEAD SOFTWARE ENGINEER, CHECKOUT BACKEND

Oct. 2020 - Oct. 2022 (2 years)

- Scala, Java, DynamoDB, Cassandra, Kubernetes, AWS · Maintained and evolved business-critical purchase systems, including a zero-downtime migration from Cassandra to DynamoDB. Owned performance, reliability and tech-debt management. Helped establish the newly formed team.

Revolut

Berlin, Germany

SENIOR SOFTWARE ENGINEER, OPEN BANKING

Jun. 2020 - Sep. 2020 (4 months)

- Java, PostgreSQL, Spring · Identified and addressed a concurrency bottleneck in the core data synchronization job, improving its reliability and performance; increased transaction enrichment accuracy.

Google

Zurich, Switzerland

SOFTWARE ENGINEER, YOUTUBE CHANNEL MEMBERSHIPS

Jul. 2018 - Mar. 2020 (1 year, 9 months)

- C++, Spanner, PubSub, CQRS, Stubby · Worked on subscription-based, user-supported YouTube monetization. Projects: iOS payments, optimized channel members retrieval latency (100x for large channels), pricing levels.
- Rust · Personal project (Google 20% time): in-place renaming of symbols in object files (ELF, Mach-O, PE/COFF) for the Mundane cryptographic library.

Tink

Stockholm, Sweden

SOFTWARE ENGINEER, CORE PLATFORM

Oct. 2016 - Jun. 2018 (1 year, 8 months)

- Java, Go, Cassandra, MySQL, Kafka, ZooKeeper, Elasticsearch, AWS, Prometheus, Grafana, Kibana, SaltStack, Terraform · Developed the core transaction-processing platform. Projects: up/down scaling, driving the move to microservices, continuous delivery pipeline, performance improvements, advocating for testing, hands-on infrastructure work.

Playtika

Kyiv, Ukraine

JAVA DEVELOPER, INFRASTRUCTURE SERVICES

Mar. 2016 - Oct. 2016 (8 months)

- Spring Boot, Apache Kafka, Couchbase · Built a service for message delivery to end users via WebSockets.
- Java, Facebook Marketing API · Built a Facebook ad-audiences management service with in-memory processing of large datasets.

Yandex

Kyiv, Ukraine

SOFTWARE ENGINEER, KINOPOISK MOVIE PLATFORM

Apr. 2014 - Mar. 2016 (2 years)

- Java, Spring Framework, Jetty, MongoDB, MySQL · Built backend integrating kinopoisk.ru (movie catalogue) with Yandex services under high load. Designed REST APIs, worked on authentication and authorization, load testing, performance optimization, JVM profiling. Facilitated team TDD adoption and mentored new project members.

DIO-Soft

Kyiv, Ukraine

JAVA DEVELOPER, STATE STREET BANK

Nov. 2013 - Mar. 2014 (5 months)

- Restlet Framework, Spring DI, JAXB, Sybase, JDBC, JUnit, Mockito · Developed a backend of a financial data monitoring service with TDD and integration testing.

NetCracker

Kyiv, Ukraine

JUNIOR SOFTWARE ENGINEER, SYSTEM INTEGRATION

Mar. 2013 - Oct. 2013 (8 months)

- Java, SOAP web services, PL/SQL, JSP, JavaScript · Delivered client integration projects.

Open source contributions

Mundane, Rust cryptographic library (<https://github.com/google/mundane>)

BUILD SYSTEM IMPROVEMENTS, ELF/MACH-O/COFF MANIPULATION

2019

- Reimplement listing of exported symbols in object files in Rust, remove dependency on Go from build system.
- Replace two-step build process with in-place symbol renaming.

goblin, Rust object file parsing library (<http://github.com/m4b/goblin>)

MACH-O, AND LIBRARY ARCHIVE PARSER IMPROVEMENTS

2019

- Prevent parser crashes on certain Mach-O files.
- Implement parsing of Windows-specific library archives.
- Implement detection of static library archive type.

Talks

Lessons Learned From Building the Fashion Store Checkout Data Platform

Berlin, Germany

ZALANDO INTERNAL

2025

Distributed coordination: ZooKeeper and Chubby

Zurich, Switzerland

GOOGLE INTERNAL

2018

Distributed Locking with MongoDB

Kyiv, Ukraine

YANDEX JAVA PARTY

2015

Education

Igor Sikorsky Kyiv Polytechnic Institute

Kyiv, Ukraine

BACHELOR OF SOFTWARE ENGINEERING

2010-2015